

# Solar System, stars, orbits and satellites

Space physics  
Paper 2 · Separate Physics only  
25 questions

Cover the answers. Answer each question out loud or on paper, then check. Tick the ones you know. Learn the 13 starred (★) questions first.

|         | QUESTION                                                                          | ANSWER                                                                             | ✓                        |
|---------|-----------------------------------------------------------------------------------|------------------------------------------------------------------------------------|--------------------------|
| 1<br>★  | Name the main objects in our Solar System.                                        | The Sun, eight planets, dwarf planets, moons, asteroids and comets                 | <input type="checkbox"/> |
| 2       | What is a galaxy?                                                                 | A huge collection of stars held together by gravity.                               | <input type="checkbox"/> |
| 3       | Give the name of our galaxy.                                                      | The Milky Way                                                                      | <input type="checkbox"/> |
| 4       | Name a cloud of dust and gas where stars form.                                    | A nebula                                                                           | <input type="checkbox"/> |
| 5<br>★  | Name the force that pulls a nebula together to form a star.                       | Gravity                                                                            | <input type="checkbox"/> |
| 6       | What is a protostar?                                                              | A hot ball of gas formed as gravity pulls a nebula together, before fusion starts. | <input type="checkbox"/> |
| 7<br>★  | Name the process that begins when a protostar becomes hot enough.                 | Nuclear fusion                                                                     | <input type="checkbox"/> |
| 8<br>★  | Explain why a main sequence star is stable.                                       | The inward pull of gravity is balanced by the outward push from fusion energy.     | <input type="checkbox"/> |
| 9       | State what a Sun-sized star becomes after its main sequence stage.                | A red giant, then a white dwarf, then a black dwarf                                | <input type="checkbox"/> |
| 10      | State what a star much bigger than the Sun becomes after its main sequence stage. | A red super giant, then a supernova, then a neutron star or a black hole           | <input type="checkbox"/> |
| 11<br>★ | State how elements up to iron are formed.                                         | By nuclear fusion in stars.                                                        | <input type="checkbox"/> |
| 12<br>★ | State how elements heavier than iron are formed.                                  | In supernovae                                                                      | <input type="checkbox"/> |

|         | QUESTION                                                                                  | ANSWER                                                                                                 | ✓                        |
|---------|-------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|--------------------------|
| 13<br>★ | State what keeps one object in orbit around another.                                      | Gravity                                                                                                | <input type="checkbox"/> |
| 14      | What is an orbit?                                                                         | The curved path of one object around another in space.                                                 | <input type="checkbox"/> |
| 15      | Name a natural satellite of Earth.                                                        | The Moon                                                                                               | <input type="checkbox"/> |
| 16      | What is an artificial satellite?                                                          | A human-made object in orbit around a planet.                                                          | <input type="checkbox"/> |
| 17<br>★ | Explain why an object in orbit is accelerating.                                           | Its direction, and so its velocity, is always changing.                                                | <input type="checkbox"/> |
| 18<br>★ | Explain why the velocity of an object in a circular orbit changes but its speed does not. | Gravity acts at right angles to its motion, changing its direction but not its speed.                  | <input type="checkbox"/> |
| 19<br>★ | State how the speed of a stable orbit depends on its radius.                              | The smaller the radius, the greater the speed.                                                         | <input type="checkbox"/> |
| 20      | State what the Sun was formed from.                                                       | A nebula (a cloud of dust and gas) pulled together by gravity.                                         | <input type="checkbox"/> |
| 21      | Name the natural satellites of planets.                                                   | Moons                                                                                                  | <input type="checkbox"/> |
| 22<br>★ | List the stages in the life cycle of a Sun-sized star.                                    | Nebula, protostar, main sequence star, red giant, white dwarf, black dwarf                             | <input type="checkbox"/> |
| 23<br>★ | List the stages in the life cycle of a star much bigger than the Sun.                     | Nebula, protostar, main sequence star, red super giant, supernova, then a neutron star or a black hole | <input type="checkbox"/> |
| 24<br>★ | State how elements made in stars are spread through the Universe.                         | By supernova explosions.                                                                               | <input type="checkbox"/> |
| 25      | What do planets orbit? What do moons and artificial satellites orbit?                     | Planets orbit the Sun.<br>Moons and artificial satellites orbit planets.                               | <input type="checkbox"/> |